SPA617

RADIO ASTRONOMY LABORATORY

Experiment 1

RF Transmission, Reflections, and Standing Waves

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1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2 Objectives

1. Determine the phase velocity and velocity factor from the frequency spacing between consecutive maxima/minima produced by reflection from an open-ended cable.
2. Study reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load.
3. Estimate $|\Gamma|$ and verify that the cable is approximately matched to $50\,\Omega$.
4. Calculate the inductance and capacitance per unit length using v_p and $Z_0 \approx 50\,\Omega$.
5. Measure the insertion loss and attenuation per unit length.

3 Theory

3.1 Transmission lines and reflections

A coaxial cable supports TEM propagation. When a travelling wave meets a load $Z_L \neq Z_0$, part of it is reflected, with voltage reflection coefficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \tag{1}$$

so that $\Gamma = +1$ (open), -1 (short) and 0 (matched). The oscilloscope at the Tee junction sees $V_{\text{osc}} = V_{\text{inc}} + V_{\text{ref}}$; as the frequency is swept, the round-trip phase of the reflected wave changes and the amplitude oscillates between a maximum and a minimum.

3.2 Phase velocity

For a cable of length ℓ and phase velocity v_p, the round-trip delay is $\tau = 2\ell/v_p$ and the round-trip phase is

$$\phi_{\text{RT}} = 2\pi f\tau + \phi_\Gamma = \frac{4\pi f\ell}{v_p} + \phi_\Gamma, \tag{2}$$

with $\phi_\Gamma = 0$ for an ideal open circuit. Consecutive maxima (or minima) differ by 2π in phase, hence $2\pi\,\Delta f\,\tau = 2\pi$, giving

$$\tau = \frac{1}{\Delta f}, \qquad v_p = 2\ell\,\Delta f, \qquad \text{VF} = \frac{v_p}{c}. \tag{3}$$

3.3 Reflection coefficient magnitude

With $V_{\max} = |V_{\text{inc}}| + |V_{\text{ref}}|$ and $V_{\min} = |V_{\text{inc}}| - |V_{\text{ref}}|$,

$$|\Gamma| = \frac{|V_{\text{ref}}|}{|V_{\text{inc}}|} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{4}$$

A strong reduction of $|\Gamma|$ when a $50\,\Omega$ load replaces the open end indicates that $Z_0 \approx 50\,\Omega$.

3.4 Inductance and capacitance per unit length

For a low-loss line, $v_p = 1/\sqrt{L'C'}$ and $Z_0 = \sqrt{L'/C'}$, so

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}. \tag{5}$$

3.5 Cable attenuation

The insertion loss is $\mathrm{IL} = P_{\mathrm{ref}} - P_{\mathrm{cable}}$ (dBm), and the attenuation per unit length is

$$\alpha = \frac{\mathrm{IL}}{\ell} \ \mathrm{dB\,m^{-1}}. \tag{6}$$

4 Apparatus

RF signal generator; digital storage oscilloscope (input set to $1\,\mathrm{M\Omega}$); spectrum analyser; RF Tee connector; coaxial cable under test (length $\ell = 0.91 \pm 0.005$ m); short reference cable; $50\,\Omega$ termination; open-circuit termination (unconnected end); adapters.

5 Experimental Setup and Procedure

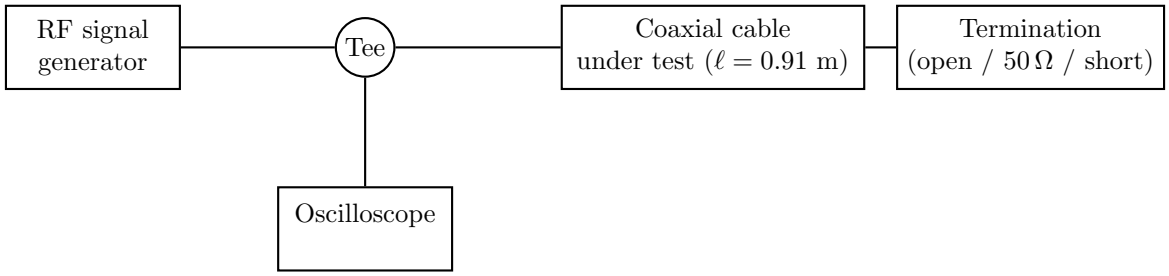


Figure 1: Schematic of the experimental setup. The generator drives the cable through an RF Tee; the oscilloscope on the side arm of the Tee records the sum of the incident and reflected waves. The far end of the cable is terminated with an open circuit, a $50\,\Omega$ load or a short.

Part A. The cable length was measured and the far end left open. A constant-amplitude sinusoid was swept slowly in frequency (3 MHz to 230 MHz) and the amplitude at the Tee was recorded; Δf was obtained from the spacing of consecutive extrema.

Part B. Maximum and minimum amplitudes were recorded with the far end open, terminated with $50\,\Omega$, and shorted, and $|\Gamma|$ was computed for each.

Part C. L' and C' were calculated from the measured v_p and $Z_0 \approx 50\,\Omega$.

Part D. Using a spectrum analyser, P_{ref} (short cable) and P_{cable} (cable inserted) were recorded at several frequencies.

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6 Observations and Calculations

6.1 Part A: phase velocity and velocity factor

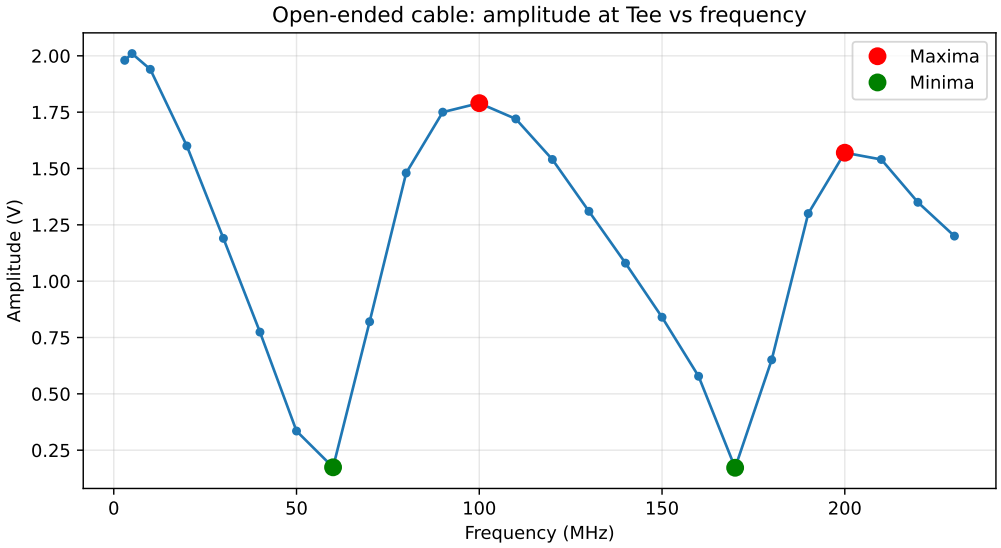


Figure 2: Measured amplitude at the Tee junction versus frequency for the open-ended cable. Maxima (red) and minima (green) used to determine Δf.

Table 1: Part A: measured amplitude at the Tee junction versus frequency for the open-ended cable. Bold entries are the extrema used for Δf (minima at 60 MHz and 170 MHz; maxima near 100 MHz and 200 MHz).

f (MHz)	V (V)	f (MHz)	V (V)
3	1.98	120	1.54
5	2.01	130	1.31
10	1.94	140	1.08
20	1.60	150	0.840
30	1.19	160	0.578
40	0.774	**170**	**0.172**
50	0.335	180	0.651
60	**0.174**	190	1.30
70	0.820	**200**	**1.57**
80	1.48	210	1.54
90	1.75	220	1.35
100	**1.79**	230	1.20
110	1.72		

The two consecutive minima (the sharpest features) lie at 60 MHz and 170 MHz, giving

$$\Delta f = 110\text{MHz}. \tag{7}$$

$$v_p = 2\ell\Delta f = 2(0.91\,\text{m})(110 \times 10^6\,\text{Hz}) = (2.00 \times 10^8)\ \text{m/s}, \tag{8}$$

$$\text{VF} = \frac{v_p}{c} = 0.667. \tag{9}$$

6.2 Part B: reflection coefficient

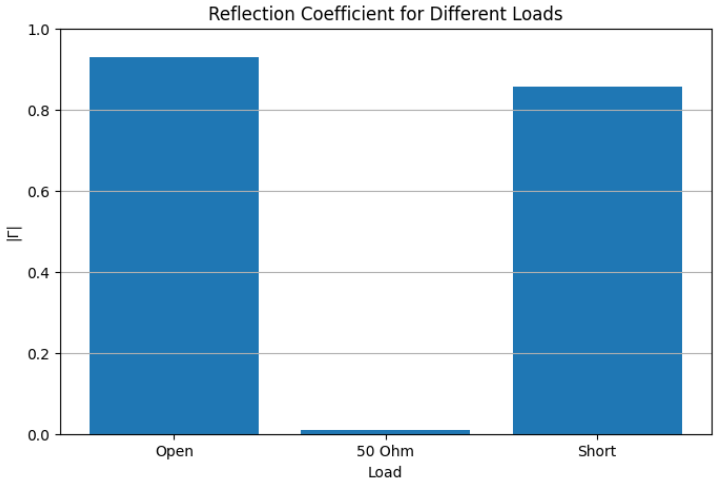


Figure 3: Reflection coefficient magnitude for the open, 50 Ω-terminated and shorted cable.

Table 2: Amplitude ripple and reflection coefficient magnitude (voltage reading uncertainty assumed ±5 mV).

Termination	$V_{\max}$ (V)	$V_{\min}$ (V)	$\lvert\Gamma\rvert$
Open circuit	0.920	0.034	0.929
50 Ω load	0.690	0.676	0.0102
Short circuit	0.988	0.076	0.857

$$|\Gamma_{\text{open}}| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = 0.929, \qquad |\Gamma_{50}| = 0.0102. \tag{10}$$

Treating the load as purely resistive ($Z_L = 50\,\Omega$), $|\Gamma_{50}| = |Z_0 - 50|/(Z_0 + 50)$ implies that Z_0 lies within 49.0–51.0 Ω, i.e. close to 50 Ω.

6.3 Part C: inductance and capacitance per unit length

Using the measured v_p and $Z_0 \approx 50\ \Omega$:

$$L' = \frac{Z_0}{v_p} = 249.8\text{nH}\,\text{m}^{-1}, \tag{11}$$

$$C' = \frac{1}{Z_0 v_p} = 99.9\text{pF}\,\text{m}^{-1}. \tag{12}$$

6.4 Part D: cable attenuation

Table 3: Measured insertion loss and attenuation of the cable ($\ell = 0.91$ m). The reference power was fixed at 0 dBm.

f (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	IL (dB)	α (dB/m)
10	0.00	-1.77	1.77	1.95
20	0.00	-1.88	1.88	2.07
30	0.00	-1.97	1.97	2.16
40	0.00	-2.06	2.06	2.26
50	0.00	-2.07	2.07	2.27
60	0.00	-2.02	2.02	2.22
70	0.00	-1.94	1.94	2.13
80	0.00	-1.88	1.88	2.07
90	0.00	-1.90	1.90	2.09
100	0.00	-2.04	2.04	2.24
150	0.00	-2.11	2.11	2.32
200	0.00	-2.53	2.53	2.78
250	0.00	-2.31	2.31	2.54
300	0.00	-2.39	2.39	2.63
350	0.00	-2.61	2.61	2.87
400	0.00	-2.47	2.47	2.71
450	0.00	-2.29	2.29	2.52
500	0.00	-2.76	2.76	3.03
700	0.00	-2.26	2.26	2.48
1000	0.00	-3.40	3.40	3.74

Mean attenuation. The mean over the $N = 20$ measured frequencies is

$$\overline{\mathrm{IL}} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{IL}_i = \frac{44.66}{20} = 2.233\,\mathrm{dB}, \tag{13}$$

$$\bar{\alpha} = \frac{\overline{\mathrm{IL}}}{\ell} = \frac{2.233}{0.91} = 2.45\ \mathrm{dB\,m^{-1}}. \tag{14}$$

Since α varies systematically with frequency, this mean is only a summary over 10 MHz to 1000 MHz; it is not a single characteristic value of the cable.

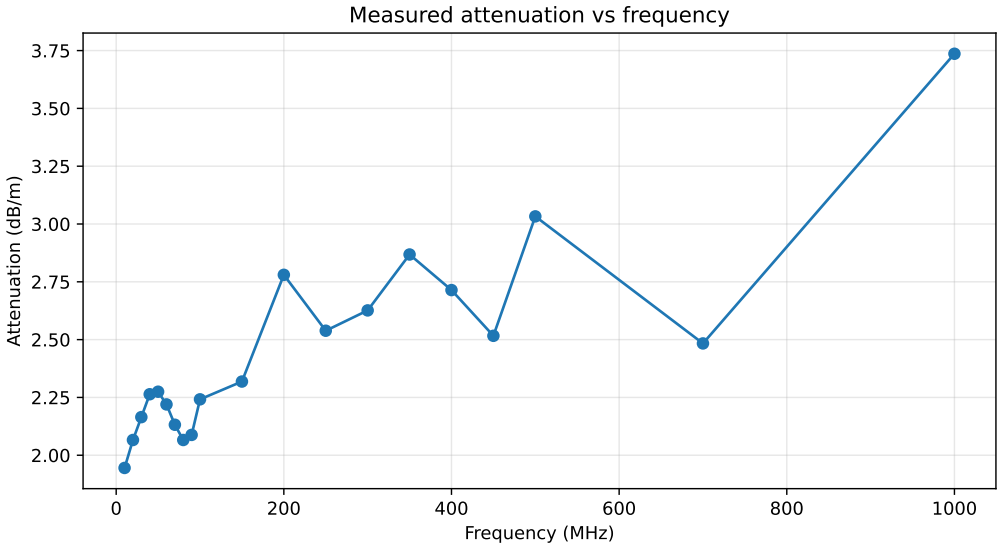


Figure 4: Measured attenuation per unit length versus frequency.

7 Results

Table 4: Summary of measured and calculated quantities.

Quantity	Symbol	Value
Cable length (m)	ℓ	0.91
Frequency spacing (MHz)	Δf	110
Phase velocity (m/s)	v_p	2.00×10^8
Velocity factor	v_p/c	0.667
Approx. characteristic impedance (Ω)	Z_0	50
Inductance per unit length (nH/m)	L'	249.8
Capacitance per unit length (pF/m)	C'	99.9
Mean attenuation (dB/m)	α	2.45

8 Conclusions

1. The amplitude at the Tee junction varies periodically with frequency due to interference between incident and reflected waves (Fig. 2).
2. Phase velocity $v_p = (2.00 \times 10^8 \pm 0.13 \times 10^8)$ m/s and velocity factor $\mathrm{VF} = 0.667 \pm 0.043$.
3. The ripple is strongly reduced with a $50\,\Omega$ termination ($|\Gamma|$: $0.929 \to 0.0102$), so the cable is approximately matched and $Z_0 \approx 50\,\Omega$.
4. $L' = 249.8 \pm 16.1$ nH/m and $C' = 99.9 \pm 6.4$ pF/m.
5. Mean measured attenuation is 2.45 dB/m over 10 MHz to 1000 MHz, increasing with frequency; this includes connector and adapter losses and so overestimates the cable's own attenuation.
6. The dominant uncertainties are the extremum-location error in Δf (about 6%) and connector repeatability in P_{cable}.